CS-E4960 - Software Testing and Quality Assurance D, Lecture, 3.9.2024-26.11.2024

This course space end date is set to 26.11.2024 **Search Courses: CS-E4960**

CI/CD Assignment 2: Configuring a Pipeline

To do: Make a submission **To do:** Receive a grade

Due: Tuesday, 26 November 2024, 10:00 AM

Objective:

In this assignment, you will configure a pipeline in Azure DevOps to automate the testing of the Django Oscar project. This will help to make sure that your project is continuously integrated and tested with every commit, enhancing the quality and reliability of your codebase.

Instructions:

1. Go to the Pipelines Tab:

- Go to your Azure DevOps project ("Oscar") by logging in to Azure DevOps with the account you created in the first assignment.
- Click on the Pipelines tab on the left-hand menu.
- Create a New Pipeline:
- While creating the pipeline, keep in mind that the code is hosted in the Azure Repos Git from the previous assignment.
- Select your "Oscar" repository.
- Configure the Pipeline:
- You can select Python Django as the template since Django Oscar is a Django-based project.
- Azure Pipelines will generate a basic YAML pipeline configuration for you. This configuration might not be perfect, so you can either:
  - Save and Run the default pipeline (it's expected to fail), or
  - Start Configuring the pipeline immediately.

2. Modify the Pipeline Configuration:

- Define three stages in your pipeline: Test, Lint, and Build.
- In each stage, set up the necessary jobs to:
  - Install the required software and dependencies.
  - Run tests, linters, and build processes.
- Testing: In the Test stage, ensure that your pipeline runs the Django Oscar tests. Consider what steps are needed to prepare the environment and execute the tests successfully.
- Linting: In the Lint stage, configure the pipeline to check your code quality. Implement jobs to lint both Python and JavaScript code in your project.
- Building: In the Build stage, automate the process of building your project's sandbox environment and generating documentation. Decide how to structure these jobs to ensure the build is successful.
- You can use the provided initial-pipeline.yaml file as a starting point for this assignment. Modify it to fit the specific needs of the Django Oscar project and assignment.

Note: If you get the error `##[error]No hosted parallelism has been purchased or granted...`, this means your request for parallelism hasn't been approved yet.

Remember that you have requested free parallelism in the last assignment. If you haven't received approval yet, you should contact us.

Submission Requirements:

- The URL of your Azure DevOps project.
- A screenshot of the pipeline run showing that it passed.
- Your pipeline configuration file, named as `studentnumber-pipeline.yaml` (replace "studentnumber" with your actual student number).

Tips:

- Make sure your pipeline is properly configured to run the tests in the tests folder.
- Don't worry if your initial pipeline run fails—debugging is part of the learning process.
- Refer to the Azure Pipelines documentation for additional help.
- **Hint for Linters:** For *Python*, you can use linters such as pylint or flake8. A pre-configured linting setup is already defined in the Makefile under the lint target, so you can use it with *make lint* command. You may review the Makefile for more details or create your own custom configurations.For *JavaScript*, there is already a defined ESLint configuration in the package.json. You can use it with npm command. Alternatively, you can run ESLint directly on the intended directory (src/oscar/static_src/oscar/js/oscar/) to avoid unrelated issues.
- For the **Test** stage, ensure that your job successfully runs all the tests available in the repository. You should aim to see output similar to the following in your pipeline logs:
  ========== 1647 passed, 1 skipped, 1926 warnings in 347.20s (0:05:47) ==========

Submission: File Submission. Submit a zip file named as *studentnumber-ci-cd.zip* (replace "studentnumber" with your actual student number). The zip file should contain the following: a screenshot of the pipeline run showing that it passed, a text file with the URL of your Azure DevOps project, and your pipeline configuration file.

Grading: The assignment is graded out of 4 points based on: the correct configuration and successful run of the pipeline in Azure DevOps, inclusion of relevant stages (Test, Lint, and Build) and their accurate execution, and clarity and completeness of the pipeline configuration, ensuring tests and linting checks are correctly set up for the Django Oscar project.

Use of AI and Collaboration with Other Students: Using AI tools to produce an answer for this assignment is not permitted. All parts of the submission must be produced by yourself (apart from files provided in the assignment). Collaboration with other students is not permitted. This is an individual assignment that you must author independently.

initial-pipeline.yaml 3 September 2024, 12:15 AM

Submission status

Attempt number	This is attempt 1.
Submission status	No submissions have been made yet
Grading status	Not marked
Time remaining	Assignment is overdue by: 106 days 3 hours
Last modified	-